

M708 HW 04

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Exercise 1

On $U_i \cap U_j \dots$

1. $\varphi_{ij} \oplus \psi_{ij}$
2. $\varphi_{ij} \otimes \psi_{ij}$
3. $(\varphi_{ij}^\top)^{-1}$
4. multiplication by $\det(\varphi_{ij})$

Exercise 3

The non-degenerate pairing exists because it exists fiberwise: for any vector space V of dimension r , we map $(v_1 \wedge \dots \wedge v_k, v_{k+1} \wedge \dots \wedge v_r) \in \bigwedge^k V \times \bigwedge^{r-k} V$ to $v_1 \wedge \dots \wedge v_r \in \bigwedge^r V = \det(V)$, and extend by linearity. This is clearly bilinear, and to see that it's nondegenerate, pick a basis $e_1, \dots, e_r$ for V . Since $a_I \neq 0$ for some I , we can always pair a nonzero $\sum_I a_I e_I \in \bigwedge^k V$ (where I runs over all k -tuples $1 \leq i_1 < \dots < i_k \leq n$) with some $e_{[r] \setminus I} \in \bigwedge^{r-k} V$ to get $\pm a_I e_1 \wedge \dots \wedge e_r \neq 0$ in $\det(V)$, since the rest of the terms vanish (since they must have two instances of e_i in the wedge product for some $i \in [r]$, by the pigeonhole principle).

Hence, we obtain a non-degenerate pairing $\bigwedge^k E \times \bigwedge^{r-k} E \rightarrow \det(E)$ of vector bundles (the above reasoning lets us define it in local trivializations, and it clearly glues together on overlaps). Since the fibers of E are finite-dimensional vector spaces, this pairing is perfect, so we obtain an isomorphism $\bigwedge^k E_x \cong \left(\bigwedge^{r-k} E_x \right)^* \otimes \det(E_x)$ fiberwise, and hence an isomorphism $\bigwedge^k E \times \bigwedge^{r-k} E \rightarrow \det(E)$ which is clearly natural with respect to bundle maps.

Exercise 4

Let $r = \text{rk}(E)$, $n = \dim(X)$, $k = \dim(E)$. Suppose we're given a global section $\sigma: X \setminus Y \rightarrow E_{X \setminus Y}$ of $E_{X \setminus Y}$. We propose to extend it to Y , that is, to find a preimage under the restriction map. We do so locally: let $\{U_i\}$ be a trivializing cover of X for E . Then on such an open set U , we can choose a trivialization of E , and by an exercise on the first HW, we can choose a chart $U \hookrightarrow \mathbb{C}^n$ with Y mapped biholomorphically onto $U \cap \mathbb{C}^k$. In this trivialization and chart, σ becomes just r holomorphic functions f_j on an open subset of $\mathbb{C}^n \setminus \mathbb{C}^k$. By assumption, we have $n - k \geq 2$, so by Hartog's theorem, each f_i extends to a holomorphic function on the open subset. Thus, pulling back to $U_i \subseteq X$, we have extended σ to U_i from $U_i \setminus Y$. This extension must agree on overlaps (since it's unique, by continuity or analytic continuation) so we obtain a well-defined extension of σ to a section of E over X . This extension is unique, also by analytic continuation or continuity, so we've found a unique preimage under restriction of every global section of $E_{X \setminus Y}$, hence restriction is a bijection $H^0(X, E) \rightarrow H^0(X \setminus Y, E_{X \setminus Y})$.

Exercise 5

We must assume X is connected. Let σ be a nontrivial global section of L , and τ a nontrivial global section of L^* . Then $\tau(\sigma)$ is a nontrivial global section of $\mathcal{O}_X$, since τ and σ both vanish on analytic subsets of X , and X is not the union of two proper analytic subsets (since it's connected). Thus, since X is compact, Liouville's theorem (and the maximum modulus principle) imply that $\tau(\sigma)$ is constant.

It's nonzero since it's nontrivial, so it's nowhere zero, so in particular σ is nowhere zero, so it provides a trivialization of L . Conversely, it's obvious that the trivial line bundle and its dual (the trivial line bundle) both admit nontrivial global sections.

Exercise 6

Since $L_1 \cong L_2$ on $X \setminus Y$, we know $L_1 \otimes L_2^*$ is trivial on $X \setminus Y$ (quick lemma: line bundles modulo isomorphism form an abelian group under $\otimes$, since well-definedness, associativity, and commutativity are clear, and the trivial bundle $\mathcal{O}_X$ is the identity, and L^* is the inverse of L , since we have fiberwise a perfect pairing $L \otimes L^* \rightarrow \mathcal{O}_X$ given by evaluation.) Hence, it has a nowhere-vanishing global section σ on $X \setminus Y$. But in any trivializing chart of $L_1 \otimes L_2^*$ over X , this is just a holomorphic function on the complement of a codimension ≥ 2 subset, so it extends (uniquely) to a function on that chart of X , and by an exercise from HW 1, this extension is nowhere-vanishing. By the uniqueness, these extensions agree on overlaps of charts, so σ extends to a nowhere-vanishing global section of $L_1 \otimes L_2^*$ over X . Therefore, $L_1 \otimes L_2^*$ is trivial, that is, $L_1 \cong L_2$.

Exercise 8

By continuity, there's only one way we can lift the $\mathbb{Z}/2\mathbb{Z}$ -action to $\text{Bl}_0(\mathbb{C}^2)$, and this is (in the usual homogeneous coordinates on the blowup of the one chart of $\mathbb{C}^2$, subject to $x_1 z_2 = x_2 z_1$)

$$([z_1 : z_2], (x_1, x_2)) \mapsto ([z_1 : z_2], (-x_1, -x_2))$$

under the action of the generator. We aim to give charts on $\text{Bl}_0(\mathbb{C}^2)/\{\pm 1\}$: let's first work in the $z_1 = 1$ chart U_1 (the other is treated exactly the same). Let $f: \mathbb{C}/\{\pm 1\} \rightarrow \mathbb{C}$ be the homeomorphism given by $[z] \mapsto z^2$ (clearly well-defined, bijective, and continuous, with continuous inverse) where $[z]$ denotes an orbit of the action of $\{\pm 1\}$ on $\mathbb{C}$. Since $U_1 \cong \mathbb{C} \times \mathbb{C}$, we have $U_1/\{\pm 1\} \cong \mathbb{C} \times \mathbb{C}/\{\pm 1\}$, and so we define the chart on U_1 to be $1 \times f$ composed with this homeomorphism. We do similarly for U_2 .

This induces a complex structure on $\text{Bl}_0(\mathbb{C}^2)/\{\pm 1\}$, since on the overlap, the transition map is the identity (we're doing the exact same thing fiberwise). Therefore, $\text{Bl}_0(\mathbb{C}^2)/\{\pm 1\}$ is a complex manifold.

Exercise 9

Let's compute this closure in charts and check that it's smooth everywhere. I want to use z coordinates as homogeneous coordinates on $\mathbb{P}^1$, so let the curve be $x_1^2 = x_2^2(x_2 + 1)$. In the $z_1 = 1$ chart in the blowup, we have coordinates z_2, x_1, x_2 subject to $x_2 = z_2 x_1$, and the closure of the preimage $\tilde{C}$ of $C \setminus \{0\}$ is, by continuity, defined by the equation $x_1^2 = x_2^2(x_2 + 1) = z_2^2 x_1^2 (z_2 x_1 + 1)$. Thus $1 = z_2^3 x_1 + z_2$ is the defining equation inside the blowup. Since this chart is clearly biholomorphic to $\mathbb{C}^2$ with coordinates z_2, x_1 , it suffices to check that the derivatives with respect to these variables of this defining function never both vanish. We compute:

$$\frac{\partial}{\partial z_2}(z_2^3 x_1 + z_2 - 1) = 3z_2^2 x_1 + 1, \quad \frac{\partial}{\partial x_1}(z_2^3 x_1 + z_2 - 1) = z_2^3$$

and these never both vanish, since the latter implies $z_2 = 0$, which implies $3z_2^2 x_1 + 1 = 1 \neq 0$.

Similarly, in the other chart, by rehomogenizing and setting $z_2 = 1$, we see that the defining equations in $\mathbb{C}^3$ are $x_1 = z_1 x_2$ and $x_1^2 = z_1^2 x_2^2 = x_2^2(x_2 + 1)$, so $z_1^2 - x_2 - 1 = 0$ in $\mathbb{C}^2$. As before, we compute

$$\frac{\partial}{\partial z_1}(z_1^2 - x_2 - 1) = 2z_1, \quad \frac{\partial}{\partial x_2}(z_1^2 - x_2 - 1) = -1$$

and the latter never vanishes, so $\tilde{C}$ is smooth in this chart too. Therefore, $\tilde{C}$ is smooth everywhere, so it's a smooth curve in $\text{Bl}_0(\mathbb{C}^2)$.

Exercise 10

Just look at the explicit formula for ω in a chart, on page 77 in Voisin's book:

$$\omega = \frac{i}{2\pi} \frac{(1 + \sum_i |z_i|^2) \sum_i dz_i \wedge d\bar{z}_i - (\sum_i \bar{z}_i dz_i) \wedge (\sum_i z_i d\bar{z}_i)}{(1 + \sum_i |z_i|^2)^2}$$

If we restrict to $z_n = 0$, then clearly we obtain the Fubini-Study form from one dimension lower.

Exercise 12

We have the explicit form in one chart (covering almost all of $\mathbb{CP}^n$)

$$\omega = \frac{i}{2\pi} \frac{(1 + \sum_i |z_i|^2) \sum_i dz_i \wedge d\bar{z}_i - (\sum_i \bar{z}_i dz_i) \wedge (\sum_i z_i d\bar{z}_i)}{(1 + \sum_i |z_i|^2)^2}$$

We change to polar coordinates in each component: $z_i = r_i e^{i\theta_i}$, so $|z_i|^2 = z_i \bar{z}_i = r_i^2$, and

$$dz_i = e^{i\theta_i} dr_i + ir_i e^{i\theta_i} d\theta_i = r_i^{-1} z_i dr_i + iz_i d\theta_i,$$

and accordingly $d\bar{z}_i = r_i^{-1} \bar{z}_i dr_i - i\bar{z}_i d\theta_i$. Thus

$$\begin{aligned} dz_i \wedge d\bar{z}_i &= (r_i^{-1} z_i dr_i + iz_i d\theta_i)(r_i^{-1} \bar{z}_i dr_i - i\bar{z}_i d\theta_i) \\ &= -ir_i^{-1} z_i \bar{z}_i dr_i \wedge d\theta_i - iz_i r_i^{-1} \bar{z}_i dr_i \wedge d\theta_i \\ &= -2ir_i dr_i \wedge d\theta_i \end{aligned}$$

and

$$\bar{z}_i dz_i = r_i^{-1} z_i \bar{z}_i dr_i + iz_i \bar{z}_i d\theta_i = r_i dr_i + ir_i^2 d\theta_i$$

Plugging all this in, we get

$$\omega = \frac{i}{2\pi} \frac{-2i (1 + \sum_i r_i^2) \sum_i r_i dr_i \wedge d\theta_i - (\sum_i r_i dr_i + ir_i^2 d\theta_i) \wedge (\sum_i r_i dr_i - ir_i^2 d\theta_i)}{(1 + \sum_i r_i^2)^2}$$

In the latter product of 1-forms, only the cross-terms survive since the square of any 1-form is 0, so we are left with $2i \sum_{i,j} r_i r_j^2 dr_i \wedge d\theta_j$. The diagonal terms here are of the form $r_i^3 dr_i \wedge d\theta_i$. In the former product of the scalar with 2-forms, the diagonal terms are also of the form $r_i^3 dr_i \wedge d\theta_i$, but of the opposite sign, so they cancel. Also cancelling the $-2i$, we are left with

$$\omega = \frac{1}{\pi} \frac{\left(\sum_i (1 + \sum_{j \neq i} r_j^2) r_i dr_i \wedge d\theta_i \right) - \sum_{i \neq j} r_i r_j^2 dr_i \wedge d\theta_j}{(1 + \sum_i r_i^2)^2}$$

Now we take this to the n th exterior power. Note that the expression above is explicitly a linear combination of terms $dr_i \wedge d\theta_j$ for $1 \leq i, j \leq n$. We might as well arrange the coefficients into a matrix:

$$M = \begin{pmatrix} (S - r_1^2)r_1 & -r_1 r_2^2 & -r_1 r_3^2 & \cdots & -r_1 r_n^2 \\ -r_2 r_1^2 & (S - r_2^2)r_2 & -r_2 r_3^2 & \cdots & -r_2 r_n^2 \\ -r_3 r_1^2 & -r_3 r_2^2 & (S - r_3^2)r_3 & \cdots & -r_3 r_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -r_n r_1^2 & -r_n r_2^2 & -r_n r_3^2 & \cdots & (S - r_n^2)r_n \end{pmatrix}$$

where $S = 1 + \sum_{i=1}^n r_i^2$. Furthermore, note that the only terms that survive in the wedge product are those coming from a permutation $\sigma \in S_n$, in the sense that we wedge together $dr_i \wedge d\theta_{\sigma(i)}$ for $i = 1, \dots, n$, since we must choose n 2-forms to wedge together, and if we pick any two in the same row or column above, they will have two of some 1-form, and hence vanish. Since 2-forms commute with one another, we just get $n!$ copies of the same term from each permutation σ , so we'll bring that coefficient out front in a moment.

From a permutation σ , the differential n -form we obtain (with some coefficient) is

$$\begin{aligned} dr_1 \wedge d\theta_{\sigma(1)} \wedge dr_2 \wedge d\theta_{\sigma(2)} \wedge \dots \wedge dr_n \wedge d\theta_{\sigma(n)} &= \text{sgn}(\sigma) dr_1 \wedge d\theta_1 \wedge dr_2 \wedge d\theta_2 \wedge \dots \wedge dr_n \wedge d\theta_n \\ &= \text{sgn}(\sigma) dV \end{aligned}$$

since swapping two $d\theta_i$ terms involves moving one of them over two 1-forms, then the other over one 1-form, for a net sign change of -1 , meaning transpositions correspond to signs. Therefore, using Leibniz's formula for the determinant, we see that in fact

$$\omega^n = \frac{n!}{\pi^n} \frac{\det(M)}{S^{2n}} dV$$

Now we must compute $\det(M)$. First, let's factor out $\prod_i r_i$, since the i th row is a multiple of r_i . Then, we can factor out $\prod_i r_i^2$, taking out r_i^2 from the i th column. We're left with:

$$N = \begin{pmatrix} S - r_1^2 & -r_2^2 & \dots & -r_n^2 \\ -r_1^2 & S - r_2^2 & \dots & -r_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ -r_1^2 & -r_2^2 & \dots & S - r_n^2 \end{pmatrix}$$

We note now that this is the sum of a simpler matrix N' , whose i th column consists of all $-r_i^2$ s, with the scalar matrix S . Since these are simultaneously diagonalizable, we obtain the eigenvalues of N by adding S to each of the eigenvalues of N' . We easily see that N' has one nonzero eigenvalue, which is $\sum_i -r_i^2 = 1 - S$, and the rest are 0. Thus, the eigenvalues of N are 1, and S with multiplicity $n - 1$. Therefore, $\det(M) = r_1 r_2 \dots r_n \cdot S^{n-1}$, so we have

$$\omega^n = \frac{n!}{\pi^n} \frac{r_1 r_2 \dots r_n \cdot S^{n-1}}{S^{2n}} dV = \frac{n!}{\pi^n} \frac{r_1 r_2 \dots r_n}{S^{n+1}} dV$$

Now, let's just compute the integral. First, we can integrate out all the $d\theta_i$ s, acquiring a factor of $(2\pi)^n$ and leaving us with

$$\int_{\mathbb{CP}^n} \omega^n = n! 2^n \int_{\mathbb{R}_{>0}^n} \frac{r_1 \dots r_n}{(1 + r_1^2 + \dots + r_n^2)^{n+1}} dr_1 \dots dr_n$$

Let's just do the dr_n integral, and substitute $u = 1 + r_1^2 + \dots + r_{n-1}^2$, so that $du = 2r_n dr_n$. Then we have

$$\begin{aligned} \int_{\mathbb{CP}^n} \omega^n &= n! 2^{n-1} \int_{\mathbb{R}_{>0}^{n-1}} r_1 \dots r_{n-1} \int_{1+r_1^2+\dots+r_{n-1}^2}^{\infty} \frac{du}{u^{n+1}} dr_1 \dots dr_{n-1} \\ &= n! 2^{n-1} \int_{\mathbb{R}_{>0}^{n-1}} r_1 \dots r_{n-1} \left[-\frac{1}{n} \frac{1}{u^n} \right]_{1+r_1^2+\dots+r_{n-1}^2}^{\infty} dr_1 \dots dr_{n-1} \\ &= (n-1)! 2^{n-1} \int_{\mathbb{R}_{>0}^{n-1}} r_1 \dots r_{n-1} \frac{1}{(1 + r_1^2 + \dots + r_{n-1}^2)^n} dr_1 \dots dr_{n-1} \end{aligned}$$

By induction, we clearly see that this reduces to the integral for the $n = 1$ case, which we already know is 1. Therefore, we conclude that

$$\int_{\mathbb{CP}^n} \omega^n = 1$$

as desired.

Exercise 13

We know that any Kähler metric ω on a compact complex manifold X of positive dimension must have nonvanishing cohomology class $[\omega] \neq 0 \in H^2(X, \mathbb{R})$, since otherwise the volume form $\omega^n/n!$ is 0. The only sphere with nonzero 2nd cohomology is S^2 , so it's the only sphere with a Kähler structure. (Also, it has one: the Fubini-Study metric on $\mathbb{CP}^1 \cong S^2$.)

Exercise 15

Vital throughout this problem will be to have the following multiplication formula in mind:

$$\begin{pmatrix} 1 & w_1 & w_2 \\ 0 & 1 & w_3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & z_1 & z_2 \\ 0 & 1 & z_3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & z_1 + w_1 & z_2 + w_3 z_1 + w_2 \\ 0 & 1 & z_3 + w_3 \\ 0 & 0 & 1 \end{pmatrix}$$

a) We put charts on X as follows: around any point $g \in G$, we will find a neighborhood U_g which is disjoint from all its Γ -translates γU_g , and thus is mapped homeomorphically onto $X = G/\Gamma$. The transition maps will automatically be holomorphic, since they're given by the action of some $\gamma \in \Gamma$, which is clearly holomorphic. This will also guarantee that the quotient is Hausdorff, since for any $g' \neq g$, at most one translate of U_g will contain g' , so if it's distinct, they're separated, and if it's the same, they're in an open subset of a Hausdorff space $\mathbb{C}^3$, so they're separated.

We carry this out by viewing G as $\mathbb{C}^3$ with the above group structure, and using the metric. If $\gamma \in \Gamma$ is the matrix given by $w_i \in \mathbb{Z} + i\mathbb{Z}$ for $i = 1, 2, 3$ and $g \in G$ is the other matrix with z_i s in $\mathbb{C}$, then we can lower-bound the distance $|\gamma g - g| = |(w_1, w_3 z_1 + w_2, w_3)|$ for $\gamma \neq 1$. If w_1 or w_3 is nonzero, then this norm is at least 1. If $w_1 = w_3 = 0$, then it reduces to $|(0, w_2, 0)|$, which is at least 1 when $w_2 \neq 0$. Therefore, it suffices to take an open ball of radius say $1/2$ around g , since its diameter is 1. Therefore, we have the desired charts on X , so it's a complex manifold.

b) By the above multiplication formula, we see that if $w_i \in \mathbb{Z} + i\mathbb{Z}$, then the projections onto the first and third coordinates both descend to the quotient by $\mathbb{Z} + i\mathbb{Z}$, so we have a well-defined holomorphic map. The fiber in G of any $([z_1], [z_3]) \in \mathbb{C}/(\mathbb{Z} + i\mathbb{Z}) \times \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$ is clearly biholomorphic to $(\mathbb{Z} + i\mathbb{Z}) \times \mathbb{C} \times (\mathbb{Z} + i\mathbb{Z})$, since the first and third coordinates are specified up to translation, and the second coordinate is free.

First, we check that all these copies of $\mathbb{C}$ are identified by the action of Γ : for any $w_1, w_3 \in \mathbb{Z} + i\mathbb{Z}$, the copy of $\mathbb{C}$ in the fiber above $([z_1], [z_3])$ parametrized by $(z_1 + w_1, z_2, z_3 + w_3)$ with $z_2 \in \mathbb{C}$ is sent to the one parametrized by $(z_1, z_2 + w_3 z_1, z_3)$ by $\gamma \in \Gamma$ the matrix given by $(w_1, 0, w_3)$. Having seen this, we check that the action of the subgroup of Γ preserving $\{(z_1, z_2, z_3) : z_2 \in \mathbb{C}\} \cong \mathbb{C}$ is just the action of $\mathbb{Z} + i\mathbb{Z}$ on $\mathbb{C}$; we see this by looking at the above formula and noting that $w_1 = w_2 = 0$, so this subgroup is isomorphic to $\mathbb{Z} + i\mathbb{Z}$, and it acts by translation in the second coordinate.

c) We know the tangent bundle of a Lie group such as G is trivial, since we can choose three linearly independent right-invariant vector fields X_1, X_2, X_3 on it. These descend to the quotient $X = G/\Gamma$, since Γ is acting on the right: suppose $g \in G$ and $g\gamma$ with $\gamma \in \Gamma$ are two representatives of the same

coset $g\Gamma$. Then since X_i is right-invariant, we have

$$(dR_\gamma)_g[X_i(g)] = (dR_\gamma \circ dR_g)_e[X_i(e)] = (dR_{g\gamma})_e[X_i(e)] = X_i(g\gamma)$$

so $d\pi_g[X_i(g)] = d\pi_{g\gamma}X_i(g\gamma)$, meaning the pushforwards induce a well-defined vector field $\tilde{X}_i$ on G/Γ . Since Γ is discrete, π is a local diffeomorphism, so the three vector fields are still linearly independent everywhere. Thus, the tangent bundle of $X = G/\Gamma$ is trivial. This works for any manifold which is the quotient of a Lie group by a discrete subgroup.